R_microbiome_analysis_final

Vienna

4/25/2020

Summary

The experiment aimed to study the response of honey bee microbiome to Chlorothalonil across host generations and to detect potential microbiome mediated effects on host phenotypes. ## Methods We reared adult bees under controlled lab conditions and inoculated them with a natural bee microbiome (start_pool). Bees were orally stressed with Chloropthalonil or no stressor (control). Three cages per treatment were used. Both, the stress-exposed and control microbiomes were transferred to the next cycle (cage to cage transfer) which was handled as before. The third cycle aims to study effects of the pre-exposed microbiome on phenotypes of naive bee hosts in comparison to control microbiomes. For that we transferred the microbiomes and let them be established in the bee hosts without any stress factor contact. We finally applied high amounts of the chemical to all bees with a control microbiome or a pre-exposed microbiome. Bee samples for 16S sequencing has been snap-frozen after cycle 1, cycle 2, cycle 3 BEFORE high stress and cycle 3 AFTER high stress. In addition, the macerated gut pools for transferring microbiome has been saved (start_microbiome as well as pool from each cage to cage transfer). We sequenced the V3-V4 region of the 16S region. Two DNA mock samples from ZymoResearch have been sequenced and named as "positive_control".

Data and metadata

Overview over all experimental variables:

Experimental variables find in metadata file:

sample_type <- microbiome_transfer or single bee *treatment* <- treatment used (control or which toxin) and additional information if microbiome transfer (e.g. Control_transfer), single bee (e.g. Control) *treatment2* <- only treatment, not indicating sample type *cage* <- cage number (three cages per treatment have been used) *treatment_cage* <- combined information of treatment and cage number (e.g. Control_2) *date* <- date of sampling during experiment *cycle* <- experimental cycle (cycle 1, cycle 2, cycle 3 before stress, cycle 3 after stress) *treatment_cycle* <- experimental cycle in combination with treatment information *treatment_cycle2*<- experimental cycle in combination with treatment plus sample type information

```
 [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
[16] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
[31] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
[46] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
[61] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
[76] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
[91] TRUE
```

```
[1] "Kingdom" "Phylum"  "Class"   "Order"   "Family"  "Genus"   "Species"
[8] "ASV"
```

```
[1] 1717
```

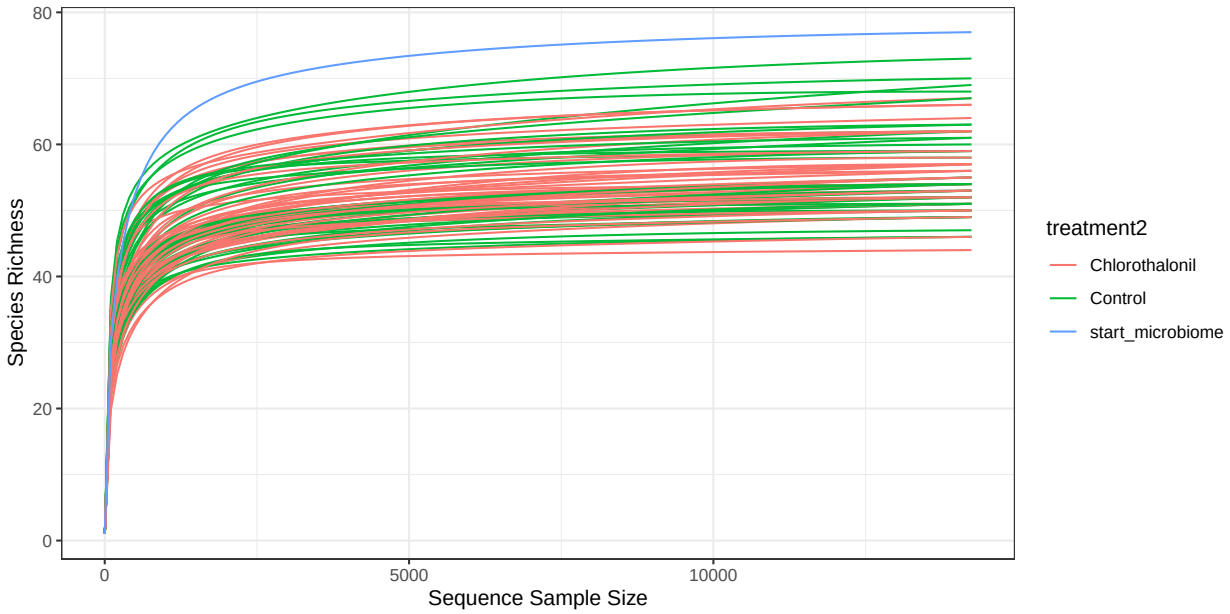
```
[1] 707

The minimum sample read count is: 14240

The average sample read count is: 27937.01

The maximum sample read count is: 49370
```

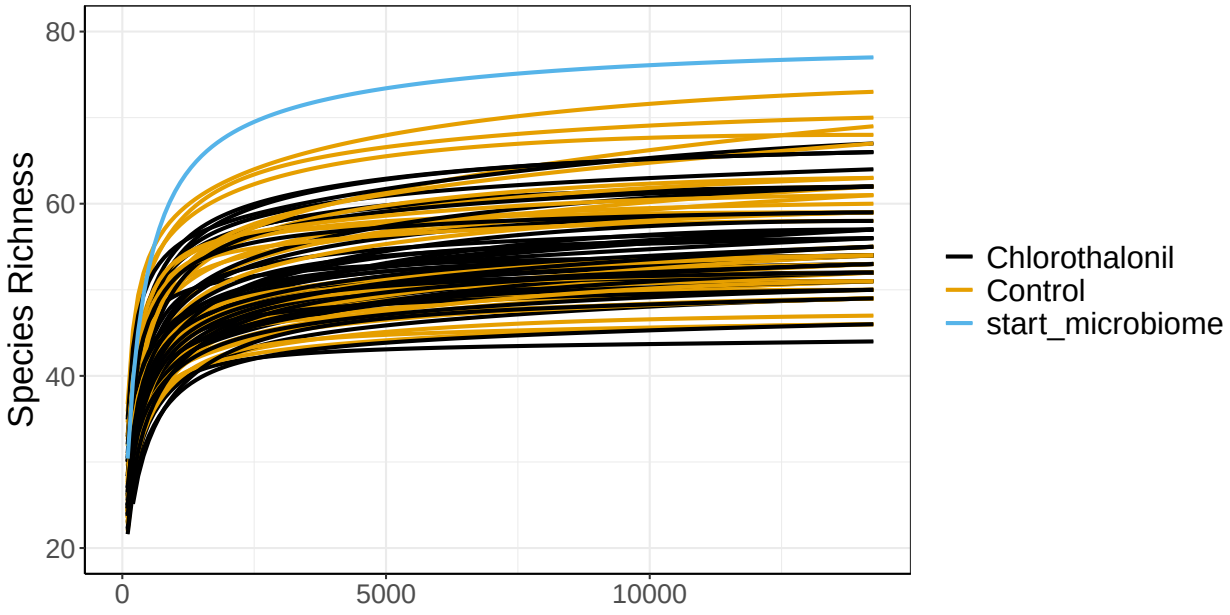
Plot rarefaction curves

```
The minimum sample read count is: 14240

phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 605 taxa and 83 samples ]
sample_data() Sample Data:       [ 83 samples by 12 sample variables ]
tax_table()   Taxonomy Table:    [ 605 taxa by 8 taxonomic ranks ]

rarefying sample X10.3.C1

rarefying sample X10.3.C2
rarefying sample X10.3.C3
rarefying sample X10.3.Ch1
rarefying sample X10.3.Ch2
rarefying sample X10.3.Ch3
rarefying sample X10.C1.1
rarefying sample X10.C1.2
rarefying sample X10.C1.6
rarefying sample X10.C2.1
rarefying sample X10.C2.4
rarefying sample X10.C2.5
rarefying sample X10.C3.2
rarefying sample X10.C3.3

rarefying sample X10.C3.4
rarefying sample X10.Ch1.1
rarefying sample X10.Ch1.2
rarefying sample X10.Ch1.3
rarefying sample X10.Ch1.4
rarefying sample X10.Ch3.1
rarefying sample X10.Ch3.2
rarefying sample X10.Ch3.3
rarefying sample X10.Ch3.4

rarefying sample X10.Ch3.5
rarefying sample X16.3.C1
rarefying sample X16.3.C2
rarefying sample X16.3.C3
rarefying sample X16.3.Ch1
rarefying sample X16.3.Ch2
rarefying sample X16.3.Ch3
rarefying sample X16.C1.4
rarefying sample X16.C1.5
rarefying sample X16.C1.8
```

```
rarefying sample X16.C2.10
rarefying sample X16.C2.5
rarefying sample X16.C2.8
rarefying sample X16.C3.2
rarefying sample X16.C3.3

rarefying sample X16.C3.7
rarefying sample X16.Ch1.4
rarefying sample X16.Ch1.5
rarefying sample X16.Ch1.9
rarefying sample X16.Ch2.2

rarefying sample X16.Ch2.6

rarefying sample X16.Ch2.8
rarefying sample X16.Ch3.2
rarefying sample X16.Ch3.4
rarefying sample X16.Ch3.6
rarefying sample X21.C1.Ch.1
rarefying sample X21.C1.Ch.2
rarefying sample X21.C1.Ch.3
rarefying sample X21.C2.Ch.1
rarefying sample X21.C2.Ch.3

rarefying sample X21.C2.Ch.6
rarefying sample X21.C3.Ch.1
rarefying sample X21.C3.Ch.4
rarefying sample X21.C3.Ch.5
rarefying sample X21.Ch1.1
rarefying sample X21.Ch1.3
rarefying sample X21.Ch1.6
rarefying sample X21.Ch2.3
rarefying sample X21.Ch2.4
rarefying sample X21.Ch2.6
rarefying sample X21.Ch3.1
rarefying sample X21.Ch3.4
rarefying sample X21.Ch3.5
rarefying sample X22.C2.Ch.1
rarefying sample X22.C2.Ch.2
rarefying sample X22.C2.Ch.3
rarefying sample X22.C3.Ch.1
rarefying sample X22.C3.Ch.2
rarefying sample X22.C3.Ch.3
rarefying sample X22.C3.Ch.4
rarefying sample X22.Ch1.1

rarefying sample X22.Ch1.2
rarefying sample X22.Ch1.3
rarefying sample X22.Ch2.1
rarefying sample X22.Ch2.2
rarefying sample X22.Ch2.3
rarefying sample X22.Ch3.3
rarefying sample X22.Ch3.5
```

```
rarefying sample X22.Ch3.6
rarefying sample X28.2.SP
```





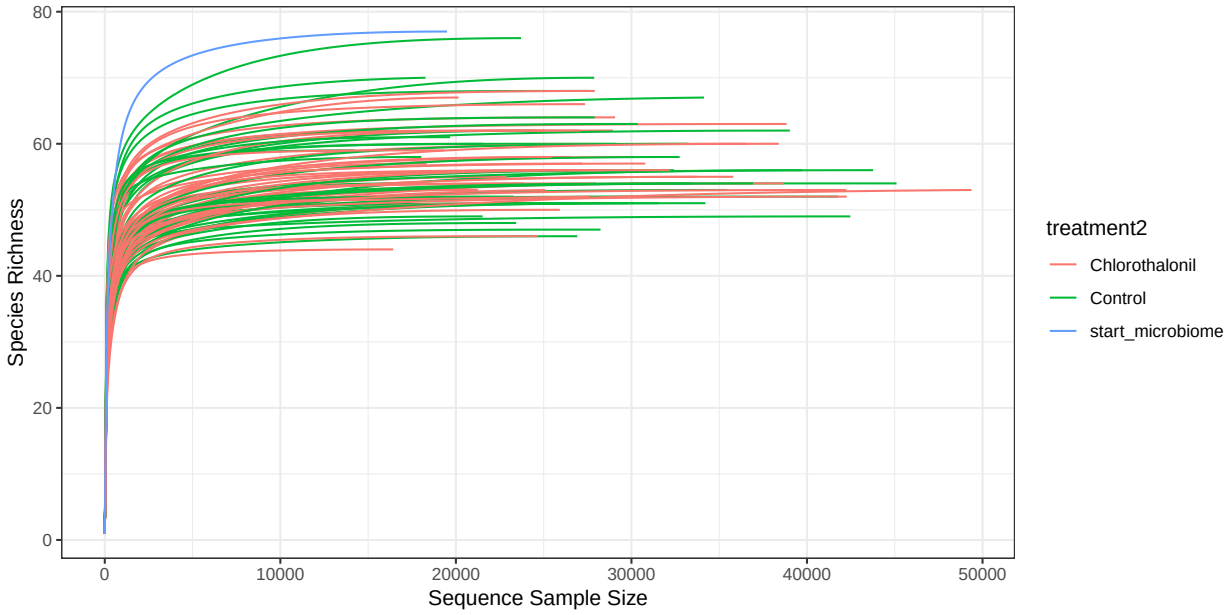

```
rarefying sample X10.3.C1

rarefying sample X10.3.C2
rarefying sample X10.3.C3

rarefying sample X10.3.Ch1

rarefying sample X10.3.Ch2

rarefying sample X10.3.Ch3
```

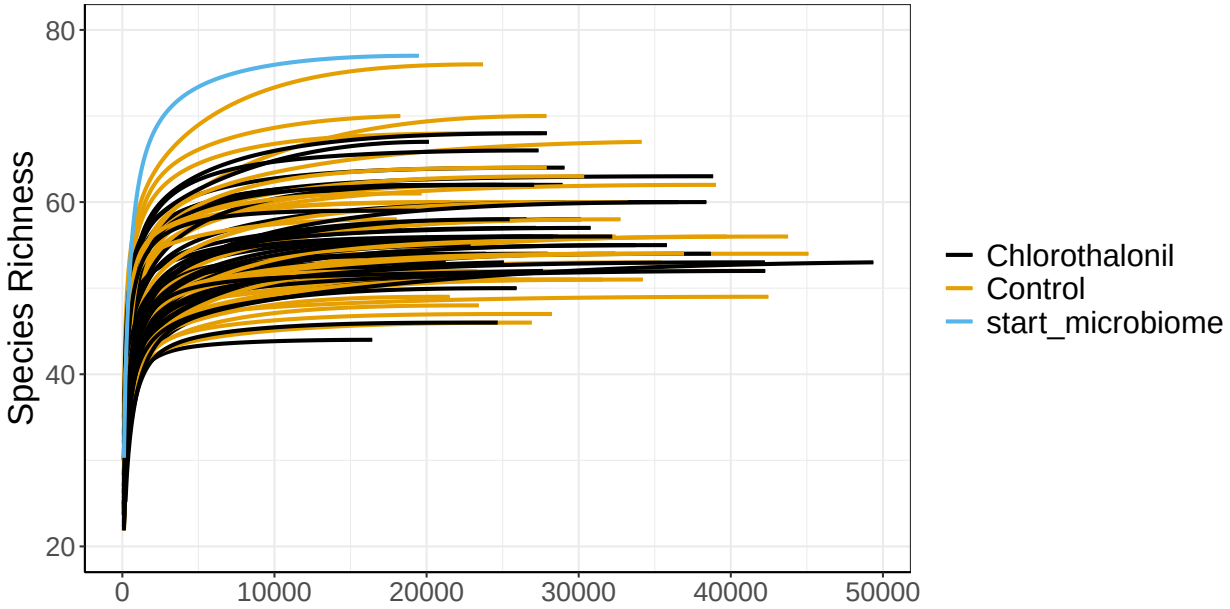
```
rarefying sample X10.C1.1

rarefying sample X10.C1.2

rarefying sample X10.C1.6
rarefying sample X10.C2.1

rarefying sample X10.C2.4

rarefying sample X10.C2.5

rarefying sample X10.C3.2

rarefying sample X10.C3.3

rarefying sample X10.C3.4

rarefying sample X10.Ch1.1

rarefying sample X10.Ch1.2

rarefying sample X10.Ch1.3

rarefying sample X10.Ch1.4

rarefying sample X10.Ch3.1

rarefying sample X10.Ch3.2

rarefying sample X10.Ch3.3

rarefying sample X10.Ch3.4

rarefying sample X10.Ch3.5

rarefying sample X16.3.C1
rarefying sample X16.3.C2

rarefying sample X16.3.C3

rarefying sample X16.3.Ch1

rarefying sample X16.3.Ch2

rarefying sample X16.3.Ch3

rarefying sample X16.C1.4
```

```
rarefying sample X16.C1.5

rarefying sample X16.C1.8

rarefying sample X16.C2.10

rarefying sample X16.C2.5

rarefying sample X16.C2.8

rarefying sample X16.C3.2

rarefying sample X16.C3.3

rarefying sample X16.C3.7

rarefying sample X16.Ch1.4

rarefying sample X16.Ch1.5

rarefying sample X16.Ch1.9

rarefying sample X16.Ch2.2

rarefying sample X16.Ch2.6

rarefying sample X16.Ch2.8

rarefying sample X16.Ch3.2

rarefying sample X16.Ch3.4

rarefying sample X16.Ch3.6

rarefying sample X21.C1.Ch.1

rarefying sample X21.C1.Ch.2

rarefying sample X21.C1.Ch.3

rarefying sample X21.C2.Ch.1

rarefying sample X21.C2.Ch.3

rarefying sample X21.C2.Ch.6

rarefying sample X21.C3.Ch.1
```

```
rarefying sample X21.C3.Ch.4

rarefying sample X21.C3.Ch.5

rarefying sample X21.Ch1.1

rarefying sample X21.Ch1.3

rarefying sample X21.Ch1.6

rarefying sample X21.Ch2.3

rarefying sample X21.Ch2.4

rarefying sample X21.Ch2.6
rarefying sample X21.Ch3.1

rarefying sample X21.Ch3.4

rarefying sample X21.Ch3.5

rarefying sample X22.C2.Ch.1

rarefying sample X22.C2.Ch.2

rarefying sample X22.C2.Ch.3

rarefying sample X22.C3.Ch.1

rarefying sample X22.C3.Ch.2

rarefying sample X22.C3.Ch.3

rarefying sample X22.C3.Ch.4

rarefying sample X22.Ch1.1

rarefying sample X22.Ch1.2

rarefying sample X22.Ch1.3

rarefying sample X22.Ch2.1

rarefying sample X22.Ch2.2

rarefying sample X22.Ch2.3
rarefying sample X22.Ch3.3
```

```
rarefying sample X22.Ch3.5
rarefying sample X22.Ch3.6

rarefying sample X28.2.SP
```





*** Supplementary figure: Rarefaction curves were used as a qualitative method to estimate the species richness as a function of sequencing depth for all samples. Rarefaction curves reached their asymptotes for all samples, suggesting that saturation in sequencing was achieved.

Positive controls / mock community

The experiment contained two positive control samples with DNA from the ZymoBIOMICS™ Microbial Community Standard (Zymo Research).

Plot taxa in Mock control samples

```
[1] 2

 [1] "Staphylococcus"       "Bacillus"             "Listeria"
 [4] "Salmonella"           "Escherichia-Shigella" "Lactobacillus"
 [7] "Enterococcus"         "Pseudomonas"          "Bartonella"
[10] "other"

# A tibble: 20 x 4
   Sample Genus                Abundance Relative_abundance
   <chr>  <fct>                    <dbl>              <dbl>
 1 PC.1   Bacillus              0.142              0.142
 2 PC.1   Bartonella            0.00145            0.00145
 3 PC.1   Enterococcus          0.0970             0.0970
 4 PC.1   Escherichia-Shigella  0.127              0.127
 5 PC.1   Lactobacillus         0.124              0.124
 6 PC.1   Listeria              0.140              0.140
 7 PC.1   Pseudomonas           0.0777             0.0777
 8 PC.1   Salmonella            0.130              0.130
 9 PC.1   Staphylococcus        0.159              0.159
10 PC.1   other                 0.000420           0.000420
11 PC.2   Bacillus              0.156              0.156
12 PC.2   Bartonella            0.00223            0.00223
13 PC.2   Enterococcus          0.106              0.106
14 PC.2   Escherichia-Shigella  0.116              0.116
15 PC.2   Lactobacillus         0.103              0.103
16 PC.2   Listeria              0.136              0.136
17 PC.2   Pseudomonas           0.0787             0.0787
18 PC.2   Salmonella            0.127              0.127
19 PC.2   Staphylococcus        0.175              0.175
20 PC.2   other                 0.000488           0.000488
```

All 8 Mock bacterial taxa correctly detected. **Figure S1: Comparison of the mock community with the theoretical proportions.** The two mock samples sequenced in this study **(A)** show no big difference to the expected theoretical proportions of the reference **(B, bar with label *theoretical*)**. The three other bacterial taxa belong to honey bee core microbiome (Bartonella, Gilliamella and Snodgrassella). The relative abundance of these three taxa across the two Mock samples account for 0.23% of all reads, representing neglectable cross-contamination during sequencing.

comparing difference between methods We sequenced two types of samples: whole bee abdomen and the mix of three macerated guts for cage to cage transfers after each cycle Therefore, we need to test if these samples are different due to the differences in the methods prior extracting before deciding if we include all or not. Use PERMANOVA on bray-curtis dissimilarities using proportional transformed abundance data

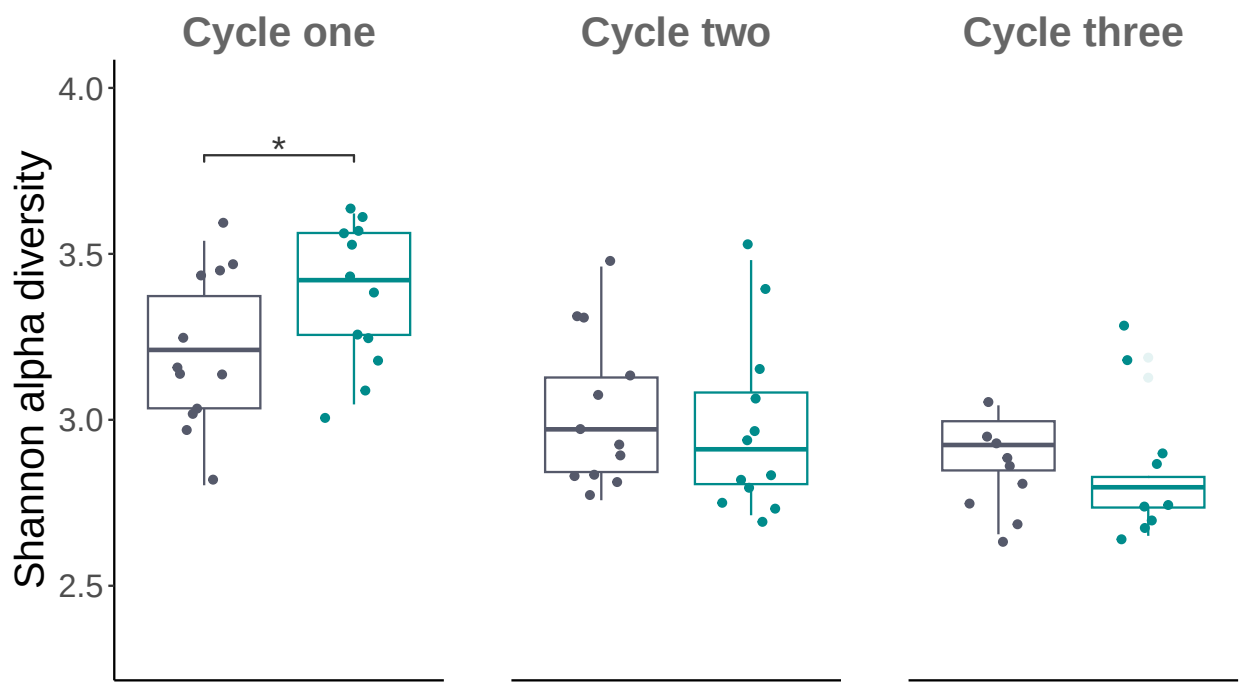
```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = phyloseq::distance(Control, method = "bray") ~ sample_type, data = C_metadata, permuta
         Df SumOfSqs      R2      F Pr(>F)
Model     1  0.10765 0.04265 1.6929  0.166
Residual 38  2.41646 0.95735
Total    39  2.52412 1.00000
```

```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = phyloseq::distance(Chloro, method = "bray") ~ sample_type, data = C_metadata, permuta
         Df SumOfSqs      R2      F Pr(>F)
Model     1   0.0715 0.02998 1.2362  0.241
Residual 40   2.3136 0.97002
Total    41   2.3851 1.00000
```

No significant difference between gut pools or whole bees and therefore, we continue with analysing them combined

Alpha diversity

```
[1] "Cycle one"   "Cycle two"   "Cycle three"
```



***The Shannon index captures information about both the species richness (number of species) and the relative abundances of the species. Increased Shannon index means increased species diversity.

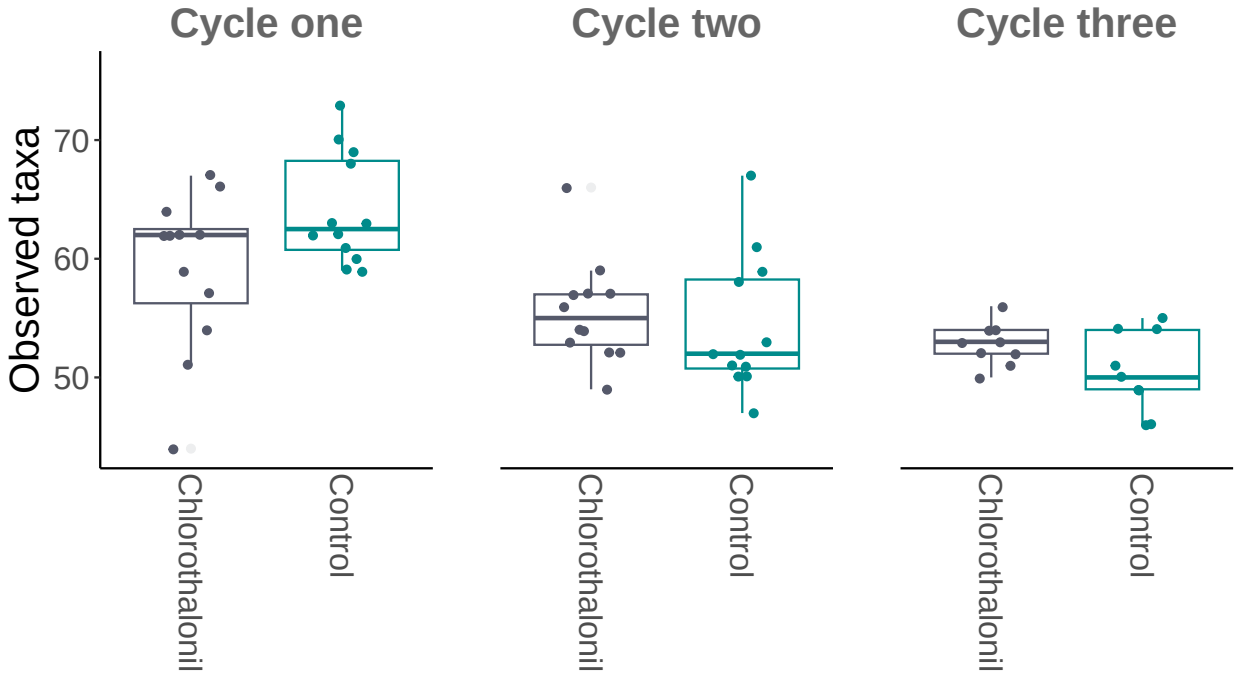
make statistical comparisons on Shannon alpha diversity index - comparing treatments against respective controls in each cycle

```
    Wilcoxon rank sum exact test

data:  Control_1 and Chlorothalonil_1
W = 106, p-value = 0.05186
alternative hypothesis: true location shift is not equal to 0
```

```
    Wilcoxon rank sum exact test

data:  Control_2 and Chlorothalonil_2
W = 63, p-value = 0.6297
alternative hypothesis: true location shift is not equal to 0


    Wilcoxon rank sum exact test

data:  control_cycle_3_before_stress and Chlorothalonil_cycle_3_before_stress
W = 30, p-value = 0.3865
alternative hypothesis: true location shift is not equal to 0
```

Under chlorothalonil the Shannon alpha diversity was significantly affected only in cycle 1

```
[1] "Cycle one"   "Cycle two"   "Cycle three"
```



*** Supplementary figure Observed alpha diversity The observed plot shows the difference in detected OTUs.

make statistical comparisons on Observed species numbers - comparing treatments against respective controls in each cycle

```
    Wilcoxon rank sum test with continuity correction

data:  Control_1 and Chlorothalonil_1
W = 99, p-value = 0.1227
alternative hypothesis: true location shift is not equal to 0
```

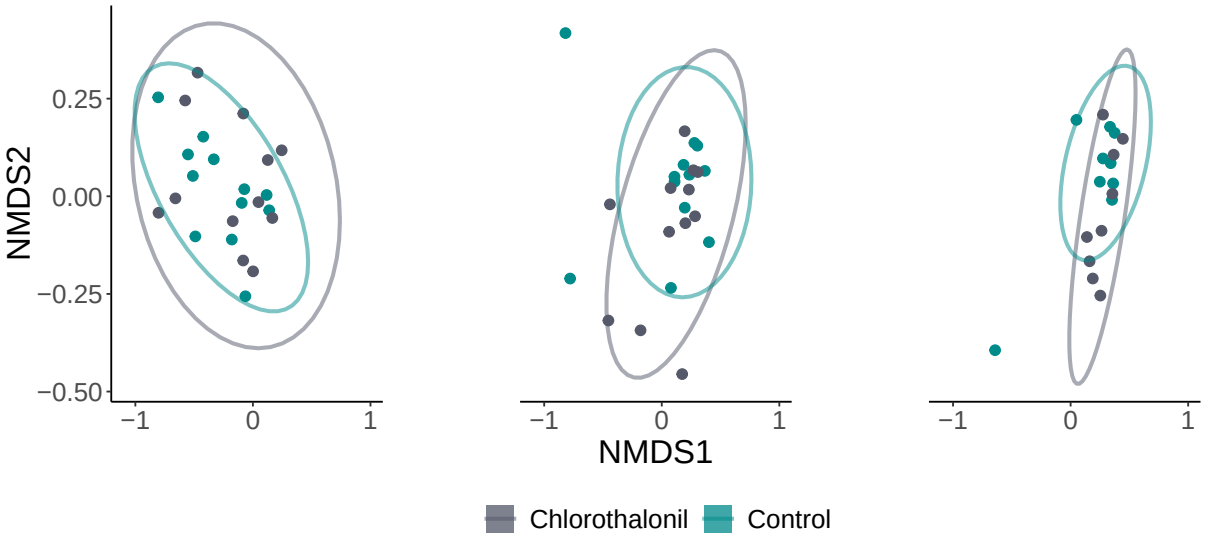
```
    Wilcoxon rank sum test with continuity correction

data:  Control_2 and Chlorothalonil_2
W = 55, p-value = 0.3388
alternative hypothesis: true location shift is not equal to 0


    Wilcoxon rank sum test with continuity correction

data:  control_cycle_3_before_stress and Chlorothalonil_cycle_3_before_stress
W = 24, p-value = 0.1543
alternative hypothesis: true location shift is not equal to 0
```

Numbers of observed species is not significantly different under chlorothalonil to the control in all three cycles.

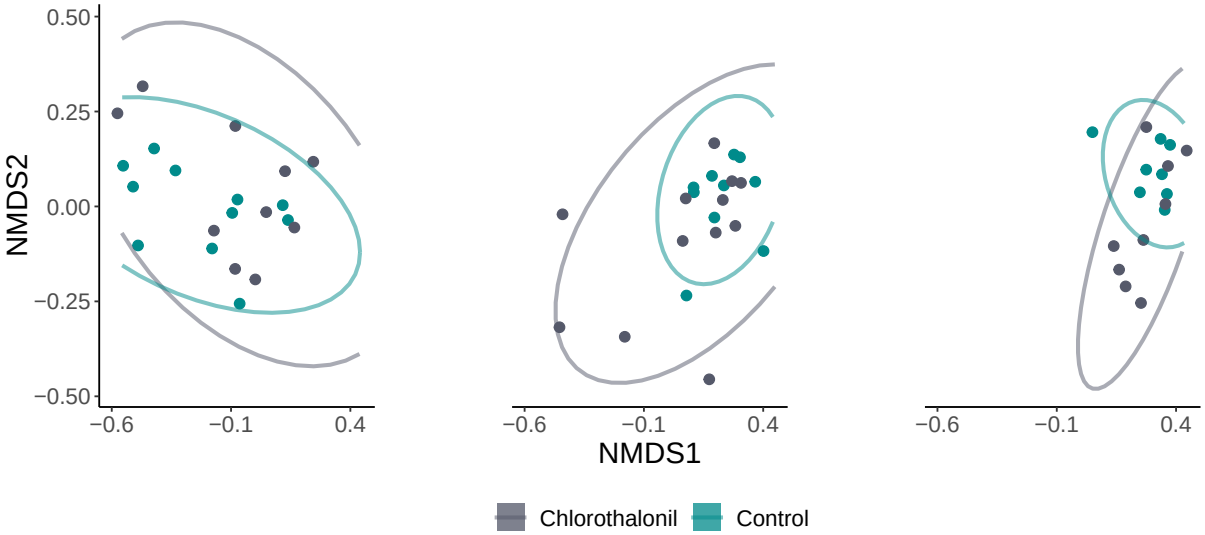
Beta diversity

ordination plots

```
[1] "Cycle one"   "Cycle two"   "Cycle three"


Run 0 stress 0.1224216
Run 1 stress 0.1425936
Run 2 stress 0.1293456
Run 3 stress 0.1347499
Run 4 stress 0.1230818
Run 5 stress 0.1303487
Run 6 stress 0.1283867
Run 7 stress 0.1358231
Run 8 stress 0.1388413
Run 9 stress 0.1368238
Run 10 stress 0.1256664
Run 11 stress 0.1303489
Run 12 stress 0.1262095
Run 13 stress 0.1286831
Run 14 stress 0.1409349
Run 15 stress 0.1228305
... Procrustes: rmse 0.01847058  max resid 0.1239126
Run 16 stress 0.1392039
Run 17 stress 0.1303486
Run 18 stress 0.1337208
Run 19 stress 0.1286021
Run 20 stress 0.1228418
... Procrustes: rmse 0.01637582  max resid 0.1251141
*** Best solution was not repeated -- monoMDS stopping criteria:
     1: no. of iterations >= maxit
    19: stress ratio > sratmax


stress is: 0.1224216
```




```
[1] "Cycle one"   "Cycle two"   "Cycle three"
```



Beta diversity stats

test difference between treatments and respective controls in the three cycles to statistically verify microbial community compositional difference seen in ordination plots

```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = phyloseq::distance(subs, method = "bray") ~ treatment2, data = metadata, permutations
         Df SumOfSqs      R2      F Pr(>F)
Model     1  0.07746 0.05636 1.3139  0.211
Residual 22  1.29692 0.94364
Total    23  1.37437 1.00000
```

```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = phyloseq::distance(subs, method = "bray") ~ treatment2, data = metadata, permutations
         Df SumOfSqs      R2      F Pr(>F)
Model     1  0.04865 0.03434 0.7824  0.497
Residual 22  1.36808 0.96566
Total    23  1.41674 1.00000
```

```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = phyloseq::distance(subs, method = "bray") ~ treatment2, data = metadata, permutations
         Df SumOfSqs      R2      F Pr(>F)
Model     1  0.06026 0.08924 1.5678  0.159
Residual 16  0.61495 0.91076
Total    17  0.67520 1.00000
```

Community composition was not significantly affected under Chlorothalonil in all cycles.

Test for cage effects

```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = d ~ cage, data = cycle1_co.1, permutations = 999)
         Df SumOfSqs      R2      F Pr(>F)
Model     2  0.09317 0.16288 0.8756  0.499
Residual  9  0.47887 0.83712
Total    11  0.57204 1.00000
```

```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = d ~ cage, data = cycle1_ch.1, permutations = 999)
         Df SumOfSqs     R2      F Pr(>F)
Model     2  0.14048 0.1938 1.0818  0.354
Residual  9  0.58439 0.8062
Total    11  0.72488 1.0000
```

```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = d ~ cage, data = cycle2_control.1, permutations = 999)
         Df SumOfSqs      R2      F Pr(>F)
Model     2  0.36945 0.47668 4.0989  0.001 ***
Residual  9  0.40560 0.52332
```

```
Total    11  0.77505 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = d ~ cage, data = cycle2_ch.1, permutations = 999)
         Df SumOfSqs      R2      F Pr(>F)
Model     2  0.11535 0.19452 1.0867  0.366
Residual  9  0.47767 0.80548
Total    11  0.59303 1.00000

Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = d ~ cage, data = cycle3_control.1, permutations = 999)
         Df SumOfSqs      R2      F Pr(>F)
Model     2  0.10893 0.28606 1.2021  0.321
Residual  6  0.27185 0.71394
Total     8  0.38078 1.00000

Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = d ~ cage, data = cycle3_ch.1, permutations = 999)
         Df SumOfSqs      R2      F Pr(>F)
Model     2  0.06693 0.28582 1.2006  0.318
Residual  6  0.16723 0.71418
Total     8  0.23417 1.00000

Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = d ~ treatment_cage, data = all1, permutations = 999)
         Df SumOfSqs    R2      F Pr(>F)
Model    14   1.3430 0.272 1.3611  0.061 .
Residual 51   3.5944 0.728
Total    65   4.9374 1.000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Testing if the three cages within a treatment show significant variations in microbiome composition (cage effects). Only in cycle 2 of the control are significant cage effects visible.

Time effect

Test if microbial composition of treatments differ across the three cycles

```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = phyloseq::distance(subs1, method = "bray") ~ cycle, data = metadata, permutations = 99
         Df SumOfSqs      R2      F Pr(>F)
Model     1  0.27332 0.16867 4.4637   0.01 **
Residual 22  1.34710 0.83133
Total    23  1.62042 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = phyloseq::distance(subs4, method = "bray") ~ cycle, data = metadata, permutations = 99
         Df SumOfSqs      R2      F Pr(>F)
Model     2  0.46568 0.23079 4.5006  0.001 ***
Residual 30  1.55207 0.76921
Total    32  2.01775 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Treatments differ significantly across cycles

compare treatments before and after high stress

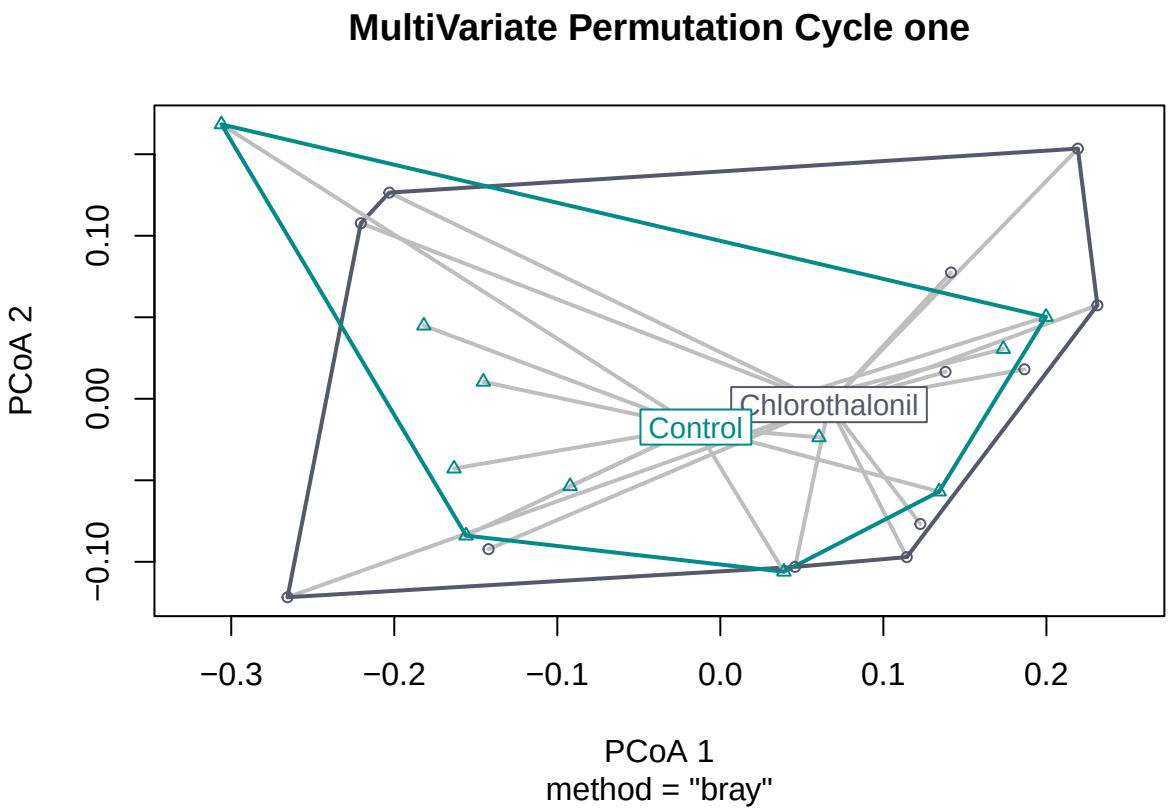
```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = phyloseq::distance(subs1, method = "bray") ~ treatment_cycle, data = metadata, permuta
         Df SumOfSqs      R2      F Pr(>F)
Model     1   0.0453 0.09021 1.3882  0.217
Residual 14   0.4568 0.90979
Total    15   0.5021 1.00000
```

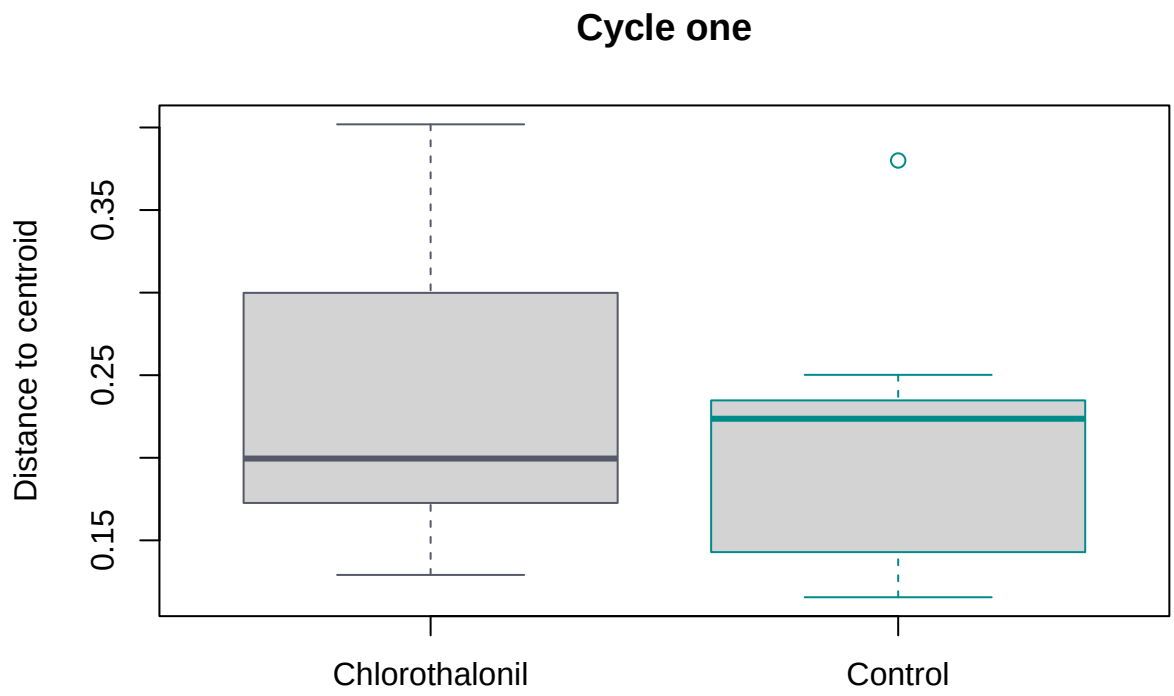
```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = phyloseq::distance(subs1, method = "bray") ~ treatment_cycle, data = metadata, permuta
         Df SumOfSqs      R2      F Pr(>F)
Model     1  0.03599 0.07484 1.2943  0.216
Residual 16  0.44489 0.92516
Total    17  0.48088 1.00000
```

No significant difference for any treatment before and after high stress application

test spread of variance of groups

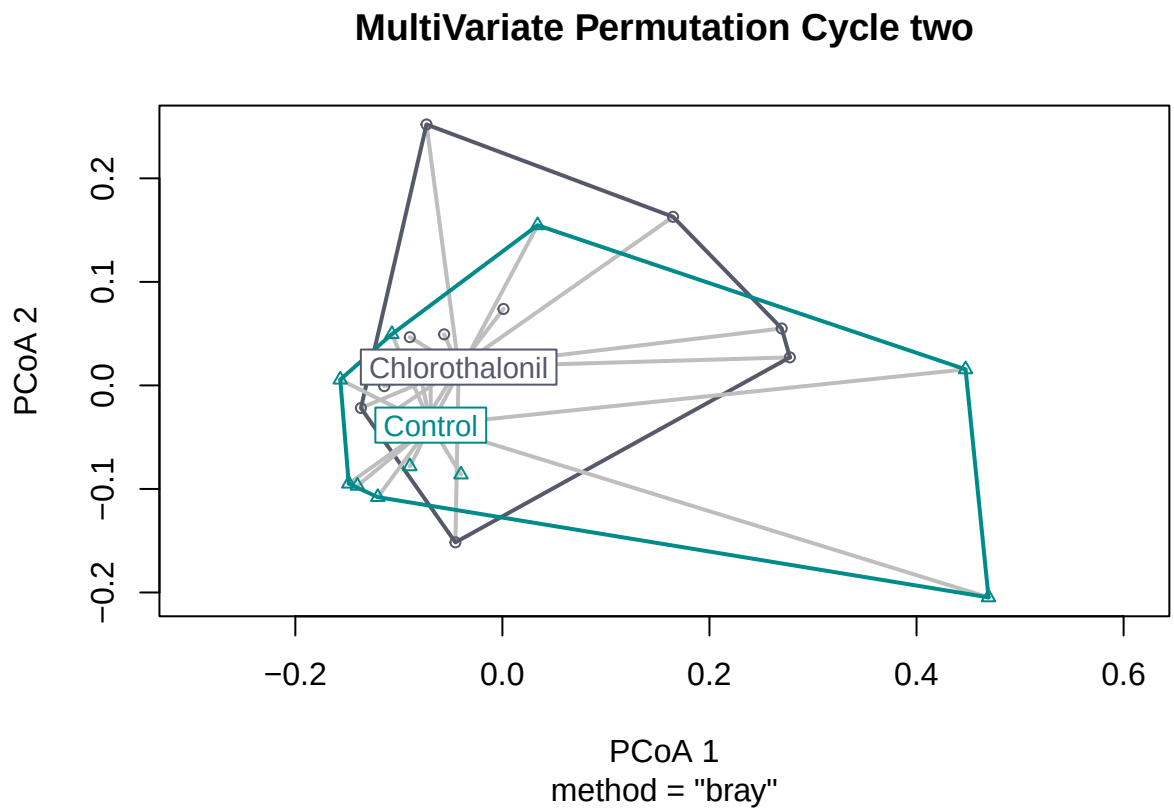
```
Permutation test for homogeneity of multivariate dispersions
Permutation: free
Number of permutations: 999

Response: Distances
          Df   Sum Sq   Mean Sq      F N.Perm Pr(>F)
Groups     1 0.004305 0.0043048 0.6823    999  0.436
Residuals 22 0.138799 0.0063090

Pairwise comparisons:
(Observed p-value below diagonal, permuted p-value above diagonal)
               Chlorothalonil Control
Chlorothalonil                  0.436
Control               0.41766

Analysis of Variance Table

Response: Distances
          Df   Sum Sq   Mean Sq F value Pr(>F)
Groups     1 0.004305 0.0043048  0.6823 0.4177
Residuals 22 0.138799 0.0063090

  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = distances ~ group, data = df)

$group
                             diff         lwr        upr     p adj
Control-Chlorothalonil -0.02678562 -0.09403493 0.04046369 0.4176553
```

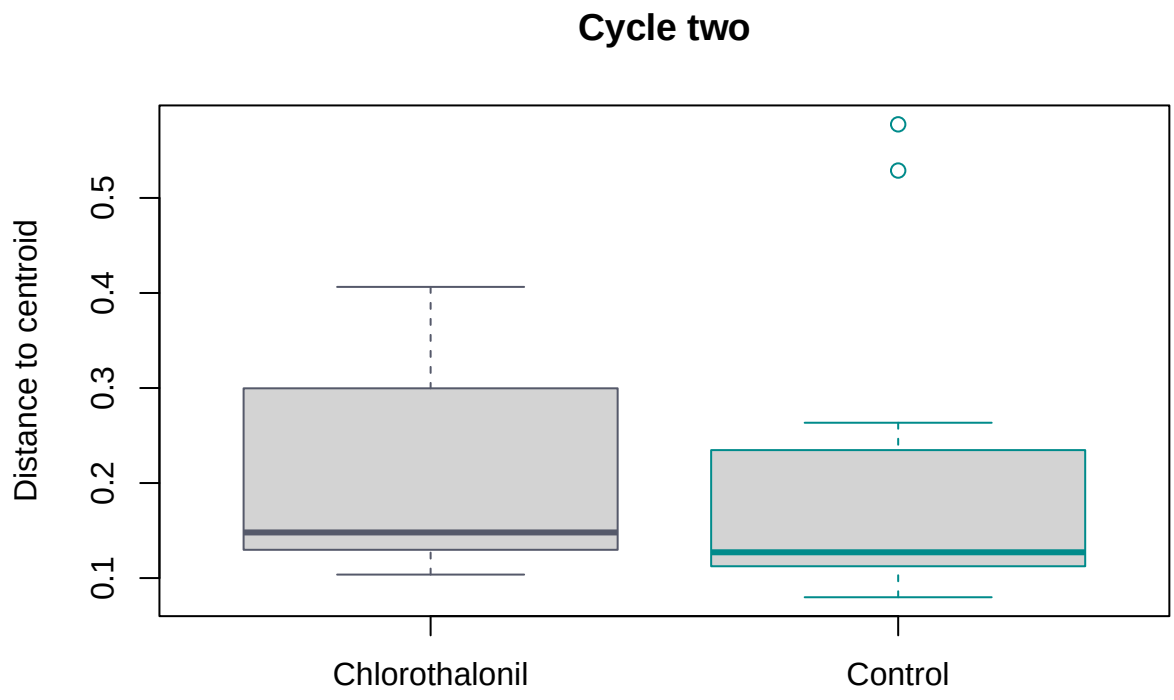

MultiVariate Permutation Cycle one
PCoA 2
0.10
0.00
−0.10
−0.3
−0.2
−0.1
0.0
0.1
0.2
PCoA 1
method = "bray"
Control
Chlorothalonil




```
Permutation test for homogeneity of multivariate dispersions
Permutation: free
Number of permutations: 999

Response: Distances
          Df  Sum Sq   Mean Sq      F N.Perm Pr(>F)
Groups     1 0.00010 0.0000997 0.0051    999  0.938
Residuals 22 0.43351 0.0197051

Pairwise comparisons:
(Observed p-value below diagonal, permuted p-value above diagonal)
               Chlorothalonil Control
Chlorothalonil                  0.944
Control               0.94393


Analysis of Variance Table

Response: Distances
          Df  Sum Sq   Mean Sq F value Pr(>F)
Groups     1 0.00010 0.0000997  0.0051 0.9439
Residuals 22 0.43351 0.0197051


  Tukey multiple comparisons of means
    95% family-wise confidence level
```

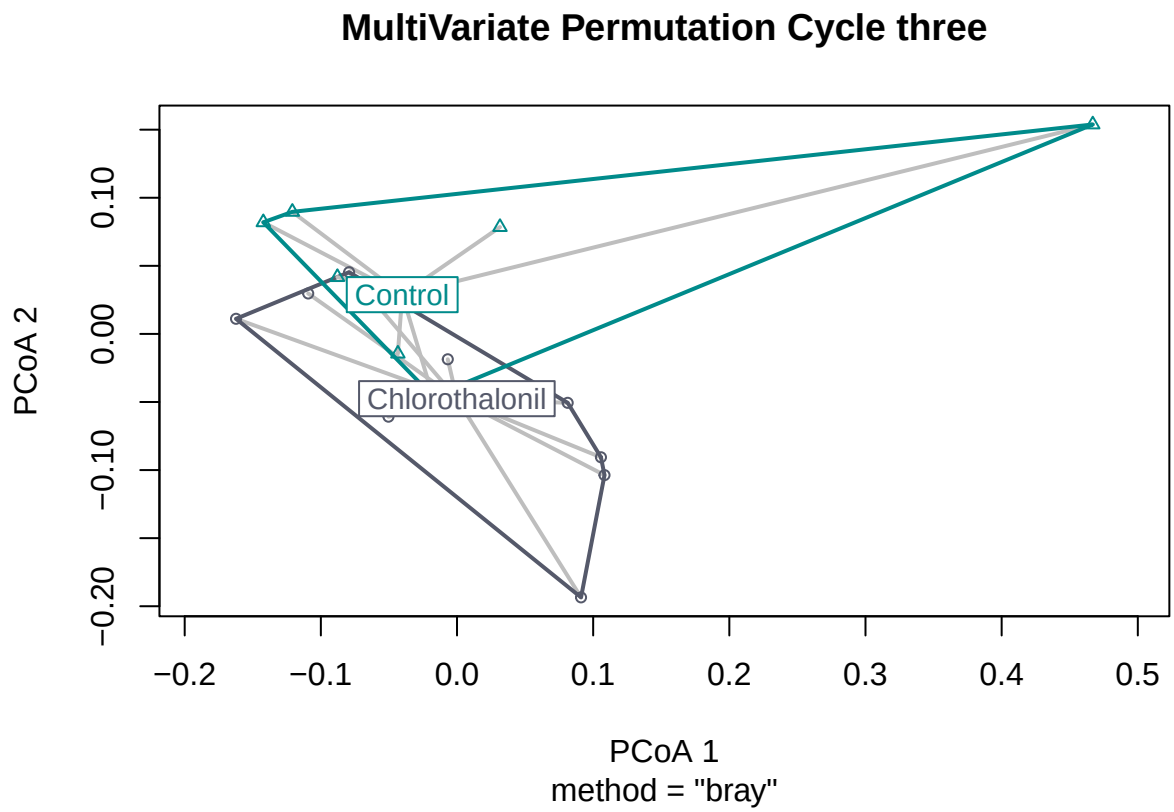
```
Fit: aov(formula = distances ~ group, data = df)

$group
                              diff        lwr       upr     p adj
Control-Chlorothalonil 0.004076469 -0.1147728 0.1229257 0.9439348
```

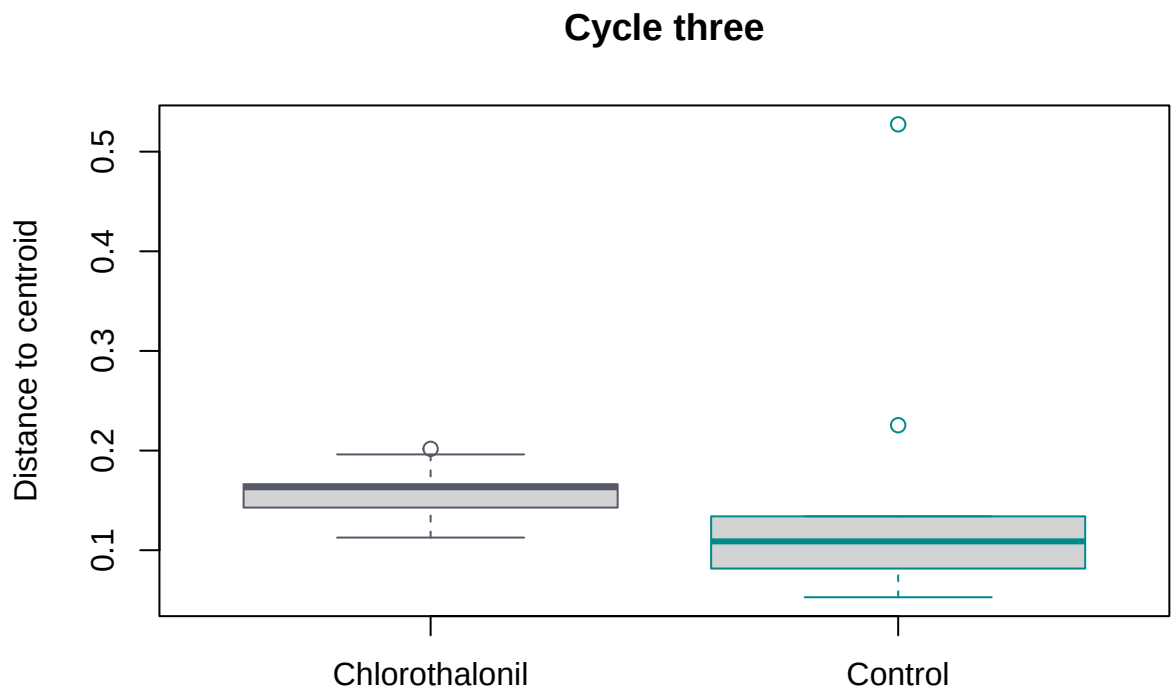





```
Permutation test for homogeneity of multivariate dispersions
Permutation: free
Number of permutations: 999

Response: Distances
          Df   Sum Sq   Mean Sq     F N.Perm Pr(>F)
Groups     1 0.000008 0.0000077 7e-04    999  0.992
Residuals 16 0.177349 0.0110843

Pairwise comparisons:
(Observed p-value below diagonal, permuted p-value above diagonal)
               Chlorothalonil Control
Chlorothalonil                  0.991
Control               0.97925


Analysis of Variance Table

Response: Distances
          Df   Sum Sq   Mean Sq F value Pr(>F)
Groups     1 0.000008 0.0000077   7e-04 0.9792
Residuals 16 0.177349 0.0110843


  Tukey multiple comparisons of means
    95% family-wise confidence level
```

```
Fit: aov(formula = distances ~ group, data = df)

$group
                              diff        lwr       upr     p adj
Control-Chlorothalonil 0.001311459 -0.1039004 0.1065233 0.9792455
```





Taxonomy plots and stats

```
[1] 19

[1] "Lactobacillus"    "Bartonella"       "Snodgrassella"    "Commensalibacter"
[5] "Gilliamella"      "Bifidobacterium"  "Frischella"       "other"
[9] "Staphylococcus"

[1] "Bartonella"       "Lactobacillus"    "Snodgrassella"    "Frischella"
[5] "Bifidobacterium"  "Gilliamella"      "Commensalibacter" "Staphylococcus"
[9] "other"

[1] "Bartonella"       "Lactobacillus"    "Snodgrassella"    "Frischella"
[5] "Commensalibacter" "Bifidobacterium"  "Gilliamella"      "other"
```

microbiome in bees cycle 3 after high stress application

```
[1] "Bartonella"       "Lactobacillus"    "Snodgrassella"    "Gilliamella"
[5] "Frischella"       "Bifidobacterium"  "Commensalibacter" "Staphylococcus"
[9] "other"
```

#nurse bees from same hive as natural control

```
[1] "Lactobacillus"    "Gilliamella"      "Bifidobacterium"  "Snodgrassella"
[5] "Frischella"       "Bartonella"       "Commensalibacter" "other"
[9] "Staphylococcus"
```

Core taxa abundance

#extract core taxa (rarefy to even depth) and plot abundance across treatments and cycles

```
# A tibble: 21 x 4
# Groups:   treatment2, Genus [7]
   treatment2 Genus         cycle       mean_val
   <chr>      <fct>         <fct>          <dbl>
 1 Control    Bartonella    Cycle one      3417
 2 Control    Bartonella    Cycle two      5643.
 3 Control    Bartonella    Cycle three    6546.
 4 Control    Lactobacillus Cycle one      5067.
 5 Control    Lactobacillus Cycle two      4355.
 6 Control    Lactobacillus Cycle three    3919.
 7 Control    Gilliamella   Cycle one       838.
 8 Control    Gilliamella   Cycle two       596.
 9 Control    Gilliamella   Cycle three     402.
10 Control    Snodgrassella Cycle one      1761.
# i 11 more rows
```

stats on abundances

compare if core taxa abundances are significantly different between controls and treatments in cycle 1, 2 and cycle 3, use Wilcoxon tests.

```
    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 58, p-value = 0.4428
alternative hypothesis: true location shift is not equal to 0
```

```
    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 83, p-value = 0.5512
alternative hypothesis: true location shift is not equal to 0
```

```
    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 41, p-value = 1
alternative hypothesis: true location shift is not equal to 0
```

```
    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 83, p-value = 0.5512
alternative hypothesis: true location shift is not equal to 0


    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 58, p-value = 0.4428
alternative hypothesis: true location shift is not equal to 0


    Wilcoxon rank sum test with continuity correction

data:  Abundance by treatment2
W = 33.5, p-value = 0.5658
alternative hypothesis: true location shift is not equal to 0
```

Gilliamella

```
    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 107, p-value = 0.0449
alternative hypothesis: true location shift is not equal to 0


    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 81, p-value = 0.6297
alternative hypothesis: true location shift is not equal to 0


    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 34, p-value = 0.6048
alternative hypothesis: true location shift is not equal to 0
```

Snodgrassella

```
    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 66, p-value = 0.7553
alternative hypothesis: true location shift is not equal to 0
```

```
    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 98, p-value = 0.1432
alternative hypothesis: true location shift is not equal to 0


    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 48, p-value = 0.5457
alternative hypothesis: true location shift is not equal to 0
```

Commensalibacter

```
    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 79, p-value = 0.7125
alternative hypothesis: true location shift is not equal to 0


    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 98, p-value = 0.1432
alternative hypothesis: true location shift is not equal to 0


    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 40, p-value = 1
alternative hypothesis: true location shift is not equal to 0
```

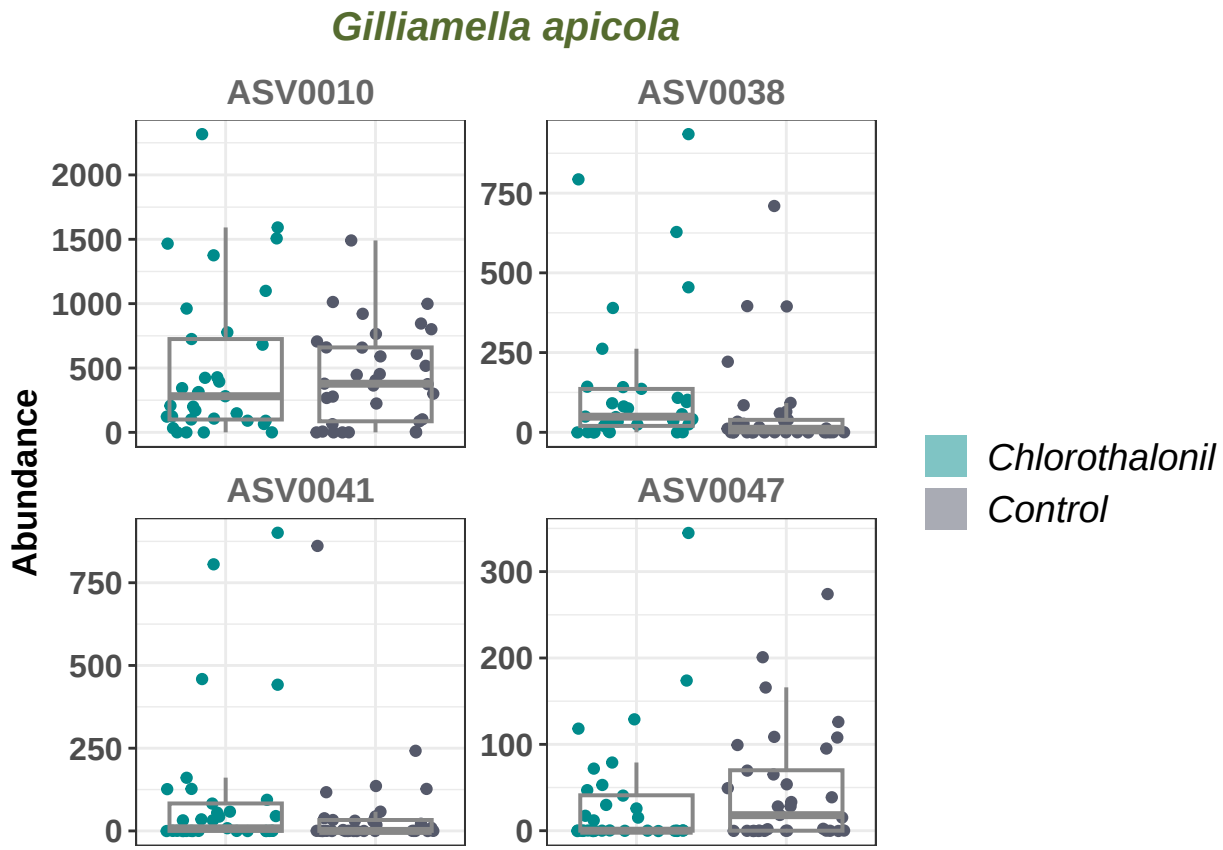
Bifidobacterium

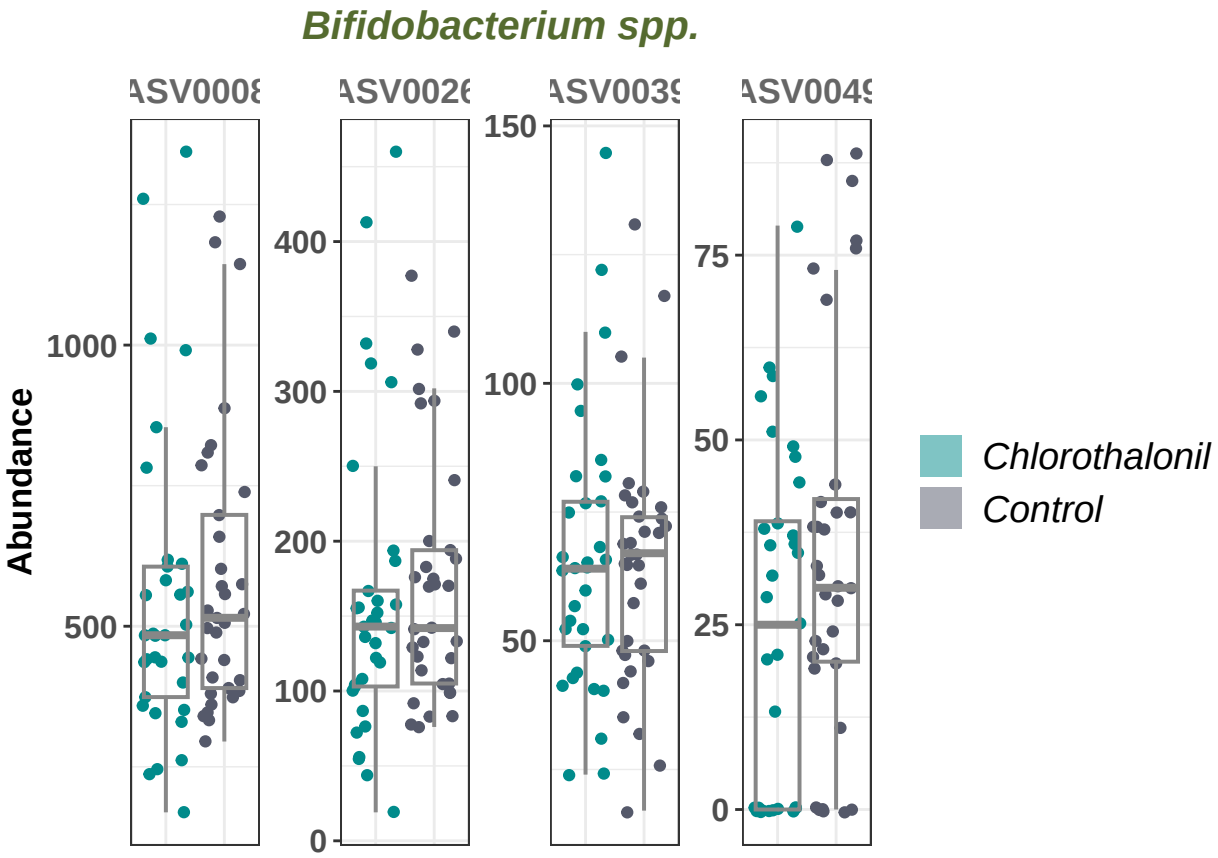
```
    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 67, p-value = 0.7987
alternative hypothesis: true location shift is not equal to 0


    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 60, p-value = 0.5137
alternative hypothesis: true location shift is not equal to 0
```

```
    Wilcoxon rank sum exact test

data:  Abundance by treatment2
W = 26, p-value = 0.2224
alternative hypothesis: true location shift is not equal to 0
```

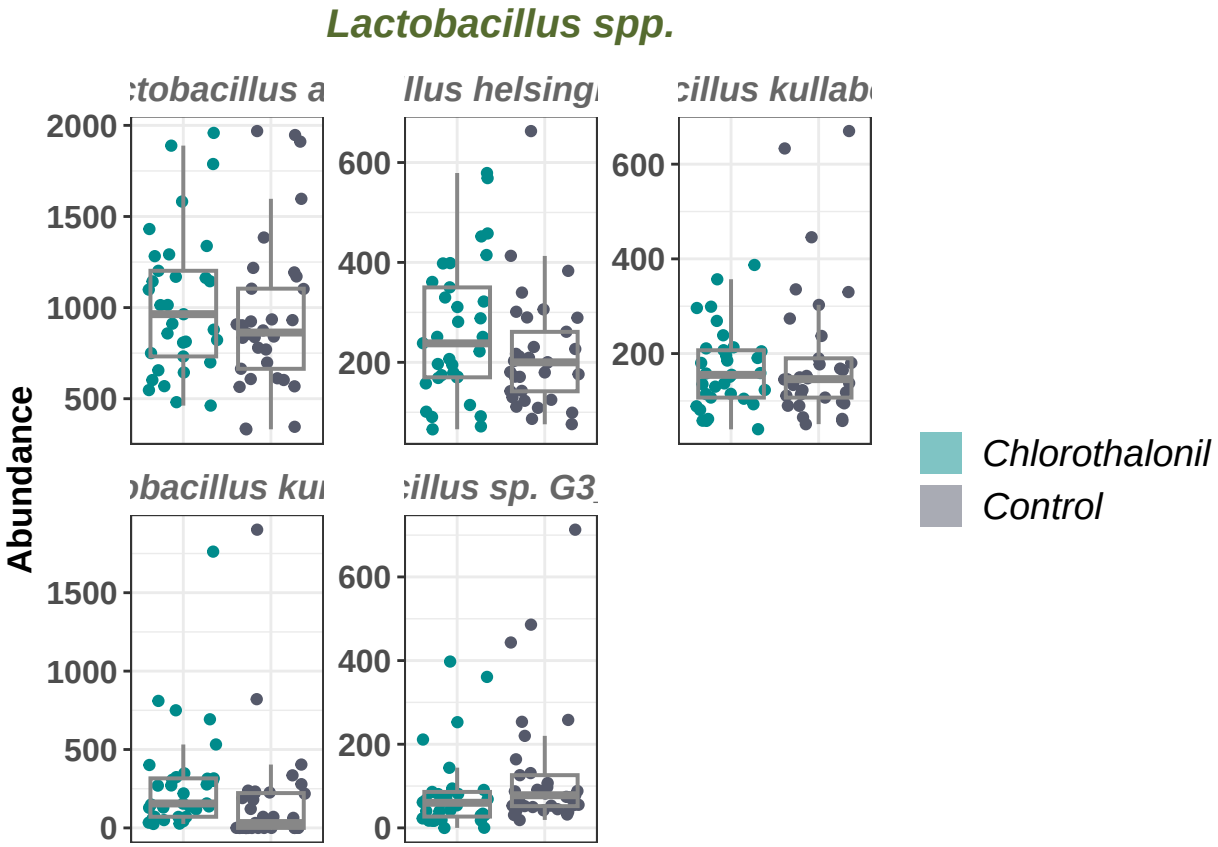
ASV

have a look on most abundant ASVs in the core taxa to test for response variation on strain level

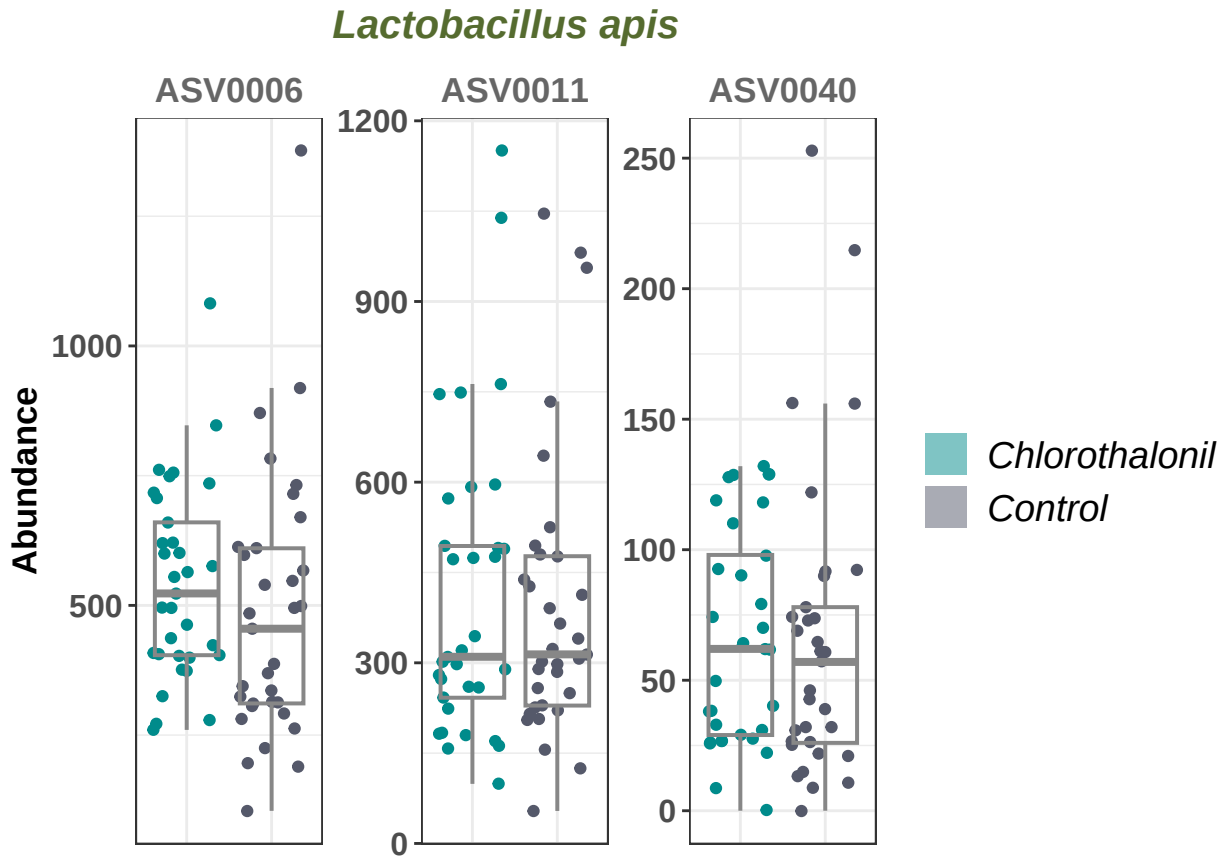


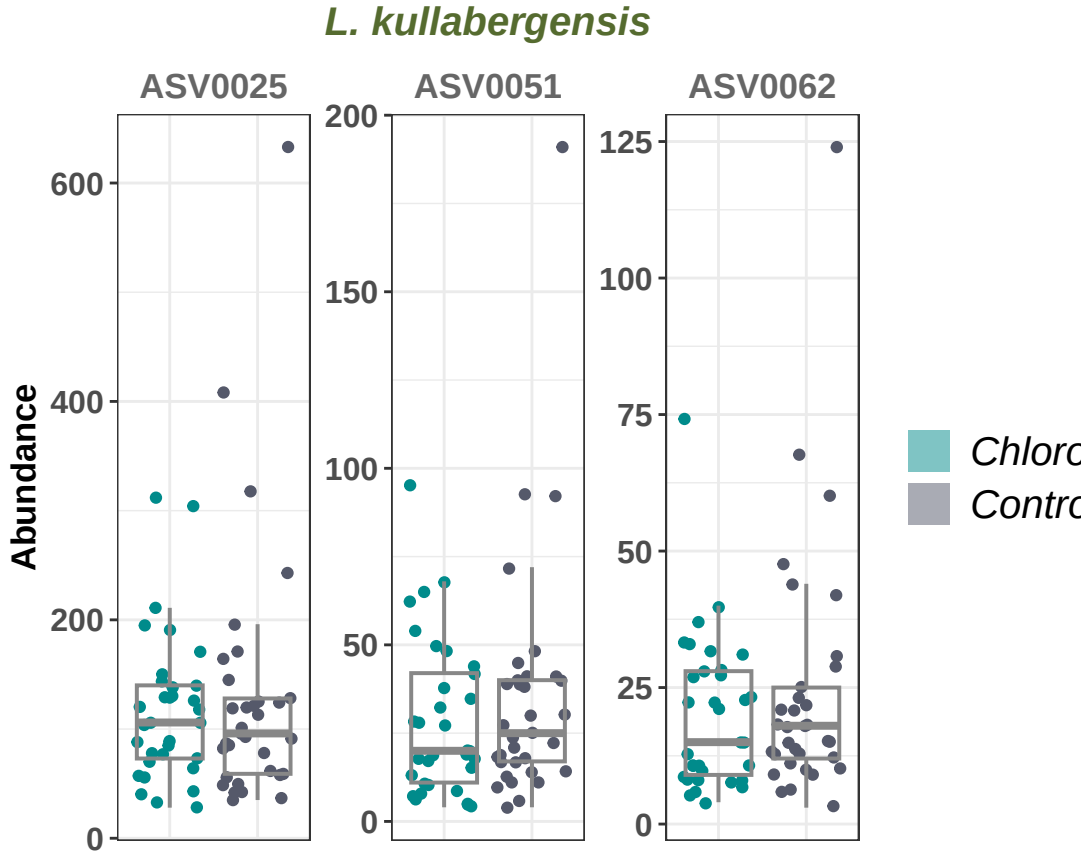

Bifidobacterium spp.
ASV0008
ASV0026
ASV0039
ASV0049
Abundance
1000
500
400
300
200
100
0
150
100
50
75
50
25
0
Chlorothalonil
Control

species Lactobacillus



ASV Lactobacillus





Lactobacillus kullabergensis